Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Design FIR filter with windowing method.**	
Experiment No: 6	**Date:24-10-2026**	**Enrollment No:92400133121**

Aim: Design FIR filter with windowing method.

Theory:

- The windowing method is a commonly used technique for designing FIR filters.
- It involves designing an ideal frequency response and then applying a window function to obtain a practical FIR filter.

The steps involved in designing an FIR filter using the windowing method are as follows:

- ➲ Specify the desired filter specifications, such as the cutoff frequency, filter length, and window type.
- ➲ Design an ideal frequency response that meets the desired specifications.
- ➲ Apply a window function to the ideal frequency response to obtain the filter coefficients.
- ➲ Normalize the filter coefficients to ensure stability.

- There are various window functions available, such as the rectangular, Bartlett, Hann, Hamming, and Blackman windows, among others.
- The choice of window function depends on the desired trade-off between the main lobe width and sidelobe suppression.

Flowchart:

The flowchart for the program will consist of the following steps:

- ❖ Specify the desired filter specifications, such as the cutoff frequency, filter length, and window type.
- ❖ Design an ideal frequency response using a mathematical expression or a frequency sampling method.
- ❖ Apply a window function to the ideal frequency response to obtain the filter coefficients.
- ❖ Normalize the filter coefficients to ensure stability.
- ❖ Plot the filter's magnitude response and impulse response.

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Code:

```
import numpy as np
import matplotlib.pyplot as plt

def design_fir_filter(cutoff_freq, filter_length, window_type='hamming'):

    M = (filter_length - 1) // 2
    n = np.arange(filter_length)

    h_ideal = 2 * cutoff_freq * np.sinc(
        2 * cutoff_freq * (n - M)
    )

    if window_type.lower() == 'hamming':
        window = np.hamming(filter_length)
    elif window_type.lower() == 'hanning':
        window = np.hanning(filter_length)
    elif window_type.lower() == 'rectangular':
        window = np.ones(filter_length)
    else:
        raise ValueError("Unsupported window type")

    h = h_ideal * window
    h = h / np.sum(h)

    return h

def plot_filter_response(filter_coefficients):

    N = 4096

    H = np.fft.rfft(filter_coefficients, n=N)
    freq = np.fft.rfftfreq(N, d=1)

    magnitude_db = 20 * np.log10(
        np.maximum(np.abs(H), 1e-10)
    )

    # Plot magnitude response
```

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```
    plt.figure(figsize=(9, 4))
    plt.plot(freq, magnitude_db)
    plt.title('FIR Filter Magnitude Response')
    plt.xlabel('Normalized Frequency (cycles/sample)')
    plt.ylabel('Magnitude (dB)')
    plt.xlim(0, 0.5)
    plt.ylim(-100, 5)
    plt.grid(True)
    plt.tight_layout()
    plt.show()

    # Plot impulse response directly from coefficients
    plt.figure(figsize=(9, 4))
    plt.stem(np.arange(len(filter_coefficients)),
            filter_coefficients)
    plt.title('FIR Filter Impulse Response')
    plt.xlabel('Sample Index')
    plt.ylabel('Amplitude')
    plt.grid(True)
    plt.tight_layout()
    plt.show()

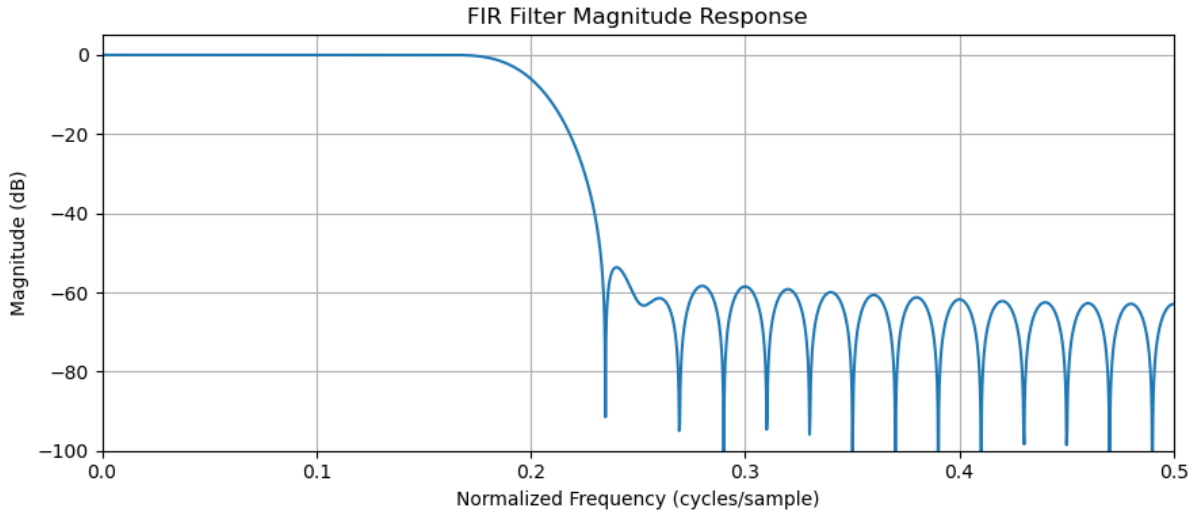
# Filter specifications
cutoff_frequency = 0.2
filter_length = 51
window_type = 'hamming'

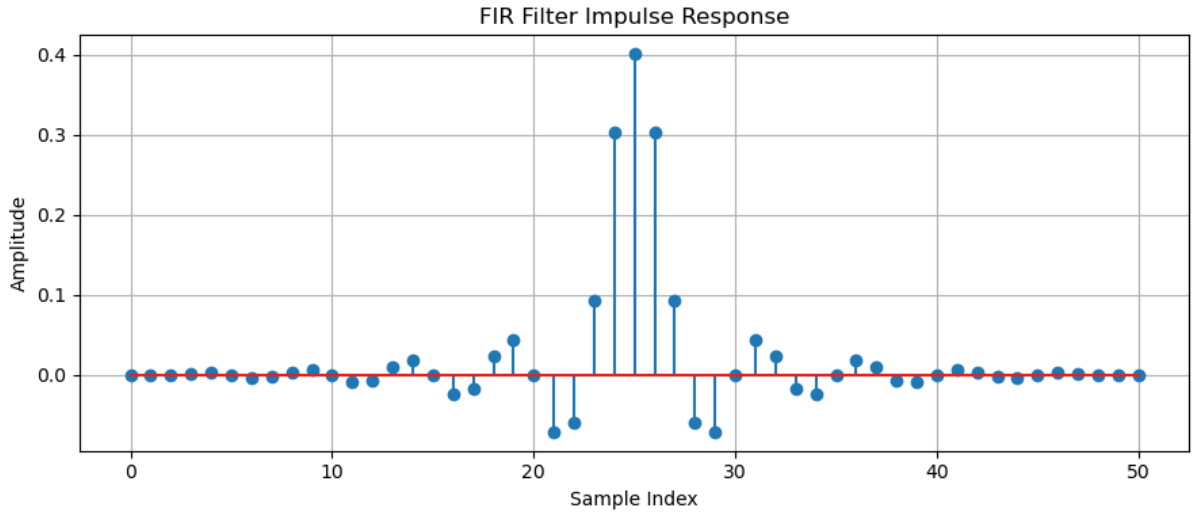
# Design and plot
filter_coefficients = design_fir_filter(
    cutoff_frequency, filter_length, window_type
)

plot_filter_response(filter_coefficients)
```

Output:

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Post Lab Work

#1. Design FIR Low pass filter having cutoff frequency 3 KHz, N=31; Sampling frequency=10 KHz

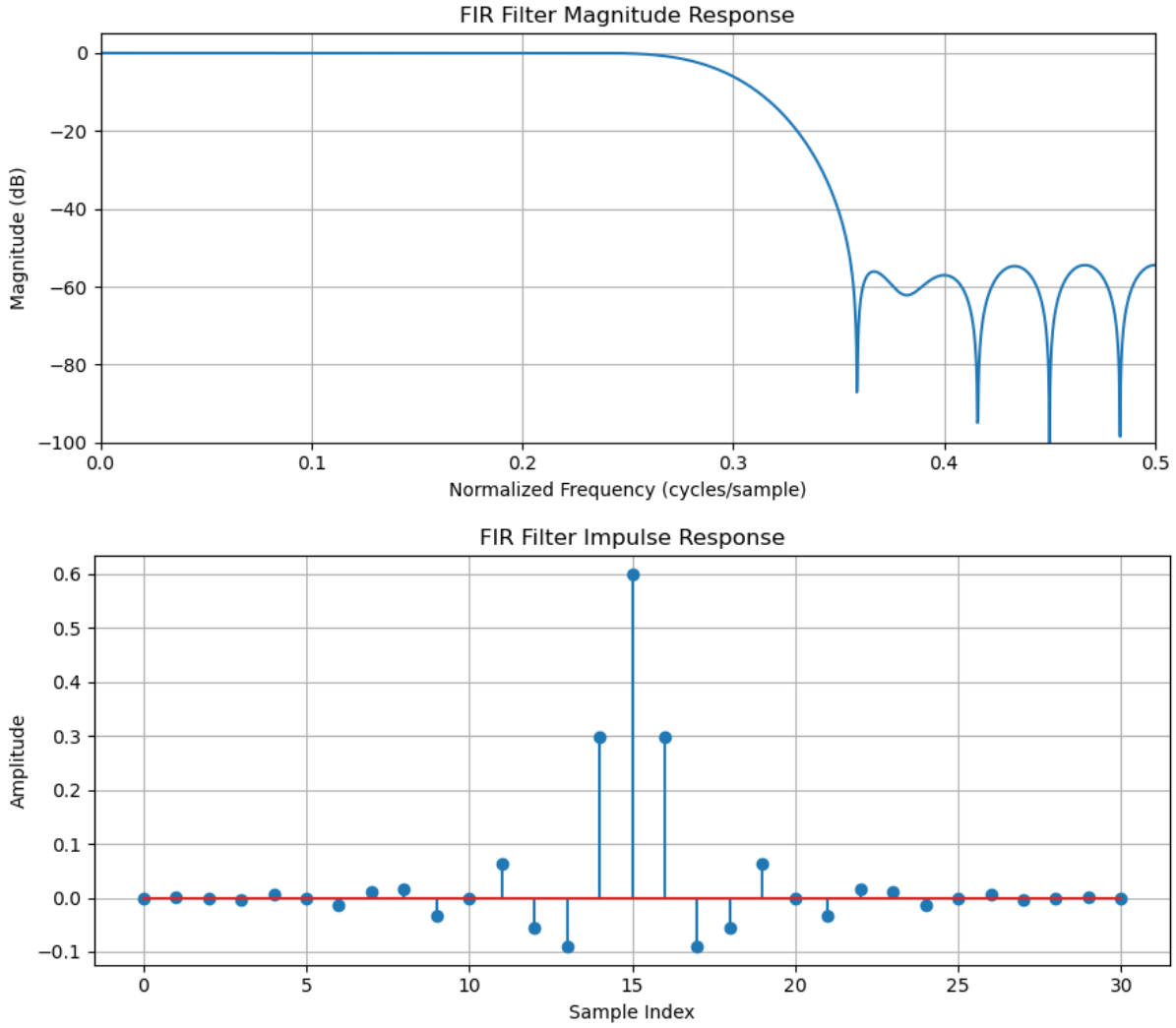
```
# Filter specifications
cutoff_frequency = 3000 / 10000
filter_length = 31
window_type = 'hamming'

filter_coefficients = design_fir_filter(cutoff_frequency, filter_length, window_type)
```

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plot_filter_response(filter_coefficients)

Output:



Observation:

- ✓ FIR low-pass filter was designed using the Hamming window method with a cutoff frequency of 3 kHz and a filter length of 31.
- ✓ The magnitude response shows that lower frequencies are passed, while frequencies above the cutoff frequency of 3 kHz are attenuated.
- ✓ The impulse response contains 31 coefficients and is symmetrical about the centre sample.

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#2. Design FIR High pass filter having cutoff frequency 3KHz, N=31Sampling frequency 10KHz

```
def design_fir_filter(cutoff_freq, filter_length, window_type='hamming'):

    M = (filter_length - 1) // 2
    n = np.arange(filter_length)

    h_ideal = -2 * cutoff_freq * np.sinc(2 * cutoff_freq * (n - M))

    h_ideal[M] += 1

    if window_type.lower() == 'hamming':
        window = np.hamming(filter_length)
    elif window_type.lower() == 'hanning':
        window = np.hanning(filter_length)
    elif window_type.lower() == 'rectangular':
        window = np.ones(filter_length)
    else:
        raise ValueError("Unsupported window type")

    h = h_ideal * window

    return h

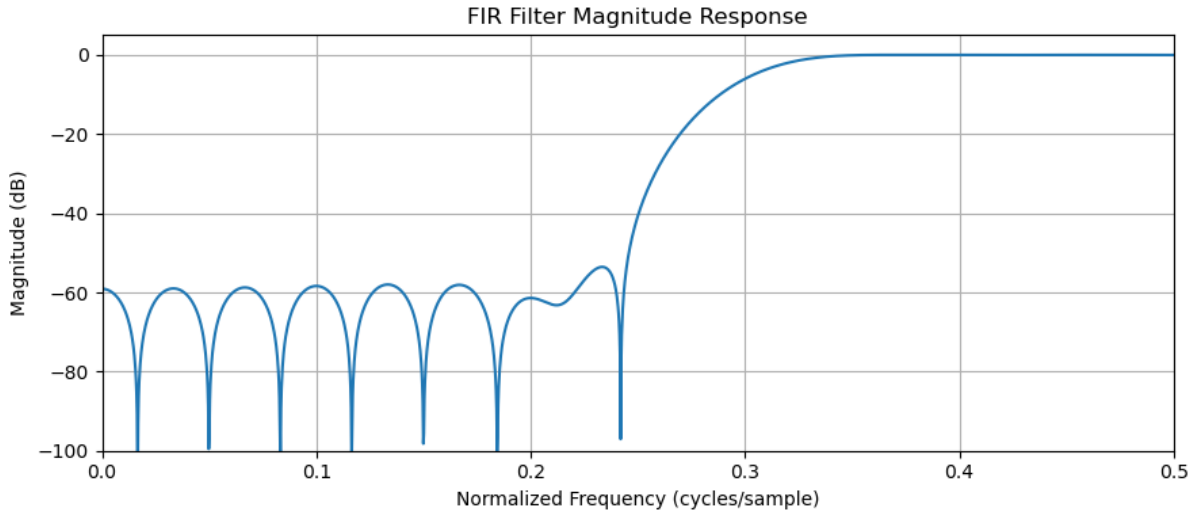
cutoff_frequency = 3000 / 10000
filter_length = 31
window_type = 'hamming'

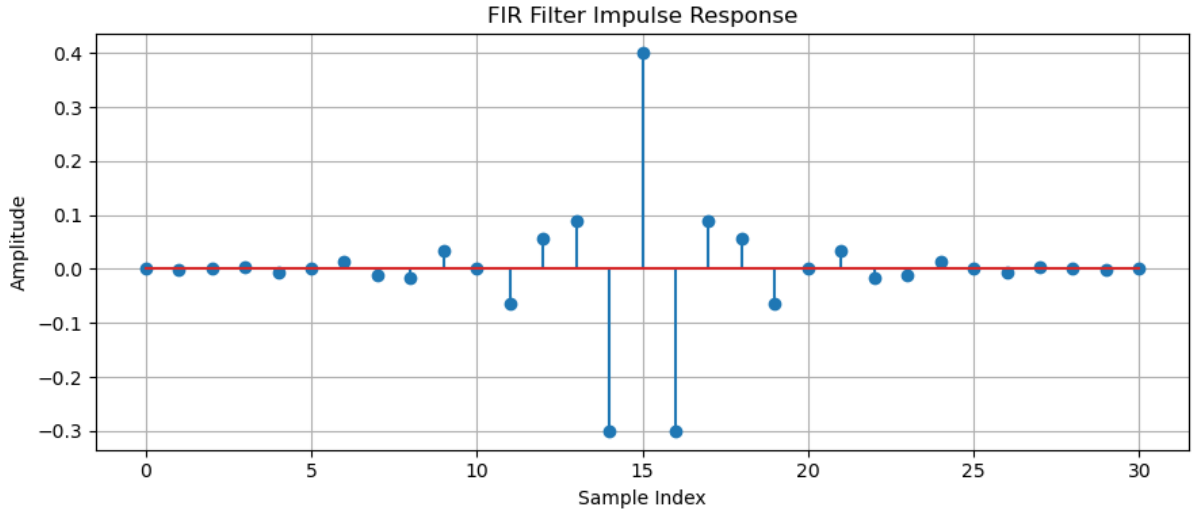
filter_coefficients = design_fir_filter(cutoff_frequency, filter_length, window_type)

plot_filter_response(filter_coefficients)
```

Output:

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Observation:

- ✓ FIR high-pass filter was designed using the Hamming window method with a cutoff frequency of 3 kHz and a filter length of 31.
- ✓ The magnitude response shows that lower frequencies are attenuated, while frequencies above the cutoff frequency of 3 kHz are passed.
- ✓ The impulse response contains 31 coefficients and is symmetrical about the centre sample.

#3. Design FIR Band pass filter having cutoff frequency (3KHz, 6KHz) N=31; Sampling frequency=10KHz

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```
def design_fir_filter(cutoff_freq1, cutoff_freq2, filter_length, window_type='hamming'):

    M = (filter_length - 1) // 2
    n = np.arange(filter_length)

    # Ideal band-pass FIR impulse response
    h_ideal = (
        2 * cutoff_freq2 * np.sinc(2 * cutoff_freq2 * (n - M))
        - 2 * cutoff_freq1 * np.sinc(2 * cutoff_freq1 * (n - M))
    )

    # Select the window
    if window_type.lower() == 'hamming':
        window = np.hamming(filter_length)
    elif window_type.lower() == 'hanning':
        window = np.hanning(filter_length)
    elif window_type.lower() == 'rectangular':
        window = np.ones(filter_length)
    else:
        raise ValueError("Unsupported window type")

    # Apply window
    h = h_ideal * window
    return h

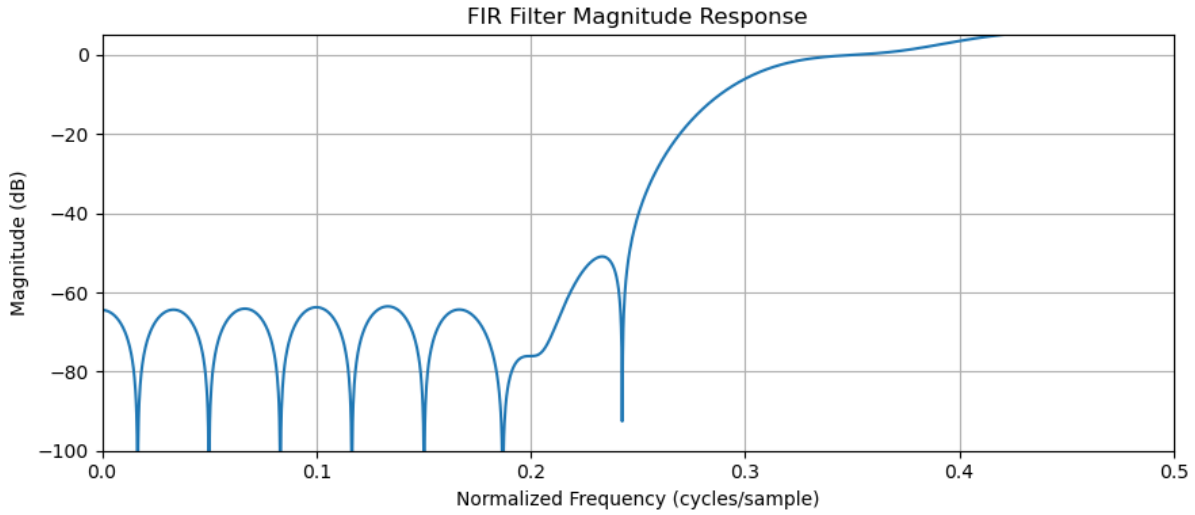
# Filter specifications
cutoff_frequency1 = 3000 / 10000
cutoff_frequency2 = 6000 / 10000
filter_length = 31
window_type = 'hamming'

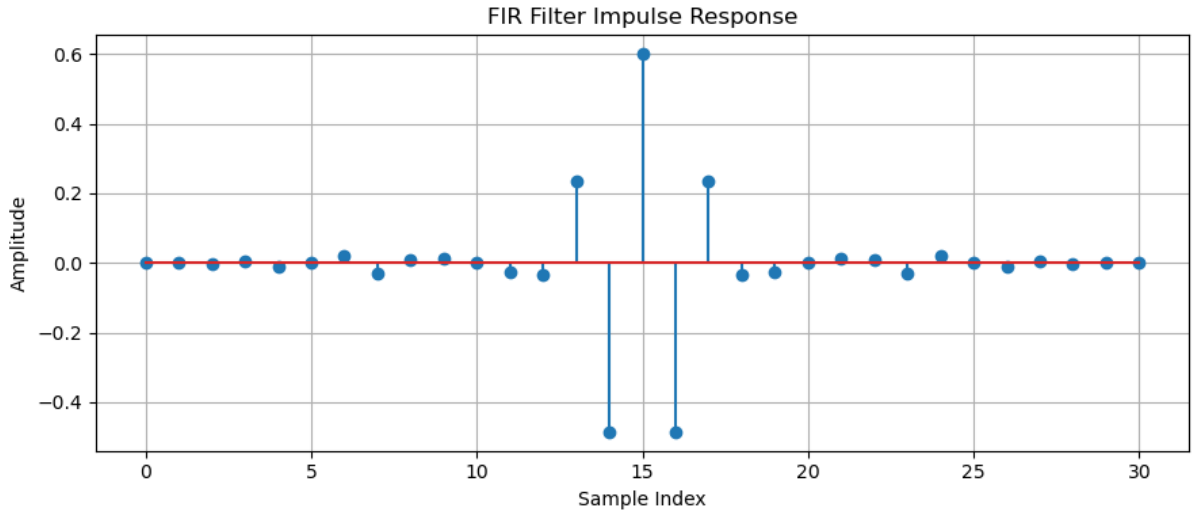
# Design and plot
filter_coefficients = design_fir_filter(
    cutoff_frequency1,
    cutoff_frequency2,
    filter_length,
    window_type
)

plot_filter_response(filter_coefficients)
```

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Output:





Observation:

- ✓ FIR band-pass filter coefficients were calculated using the windowing method with cutoff frequencies of 3 kHz and 6 kHz and a filter length of 31.
- ✓ The magnitude response was plotted to observe the frequency characteristics of the designed filter. However, the intended 3–6 kHz passband cannot be achieved because the upper cutoff frequency exceeds the Nyquist frequency of 5 kHz.

- ✓ The impulse response contains 31 coefficients and is symmetrical about the centre sample.

#4. Design FIR Band stop filter having cutoff frequency(3 KHz, 6 KHz) N=31; Sampling frequency=10 KHz

```
def design_fir_filter(cutoff_freq1, cutoff_freq2,
                      filter_length, window_type='hamming'):

    M = (filter_length - 1) // 2
    n = np.arange(filter_length)

    # Ideal band-stop FIR impulse response
    h_ideal = (
        2 * cutoff_freq1 * np.sinc(2 * cutoff_freq1 * (n - M))
        - 2 * cutoff_freq2 * np.sinc(2 * cutoff_freq2 * (n - M))
    )

    h_ideal[M] += 1

    # Select the window
    if window_type.lower() == 'hamming':
        window = np.hamming(filter_length)
    elif window_type.lower() == 'hanning':
        window = np.hanning(filter_length)
    elif window_type.lower() == 'rectangular':
        window = np.ones(filter_length)
    else:
        raise ValueError("Unsupported window type")

    # Apply window
    h = h_ideal * window

    return h

def plot_filter_response(filter_coefficients):

    N = 4096

    # Frequency response using FFT
```

```
    H = np.fft.rfft(filter_coefficients, n=N)
    freq = np.fft.rfftfreq(N, d=1)

    # Magnitude response in dB
    magnitude_db = 20 * np.log10(
       np.maximum(np.abs(H), 1e-10)
    )

    # Plot magnitude response
    plt.figure(figsize=(9, 4))
    plt.plot(freq, magnitude_db)
    plt.title('FIR Band-Stop Filter Magnitude Response')
    plt.xlabel('Normalized Frequency (cycles/sample)')
    plt.ylabel('Magnitude (dB)')
    plt.xlim(0, 0.5)
    plt.ylim(-100, 5)
    plt.grid(True)
    plt.tight_layout()
    plt.show()

    # Plot impulse response
    plt.figure(figsize=(9, 4))
    plt.stem(np.arange(len(filter_coefficients)),
           filter_coefficients)
    plt.title('FIR Band-Stop Filter Impulse Response')
    plt.xlabel('Sample Index')
    plt.ylabel('Amplitude')
    plt.grid(True)
    plt.tight_layout()
    plt.show()

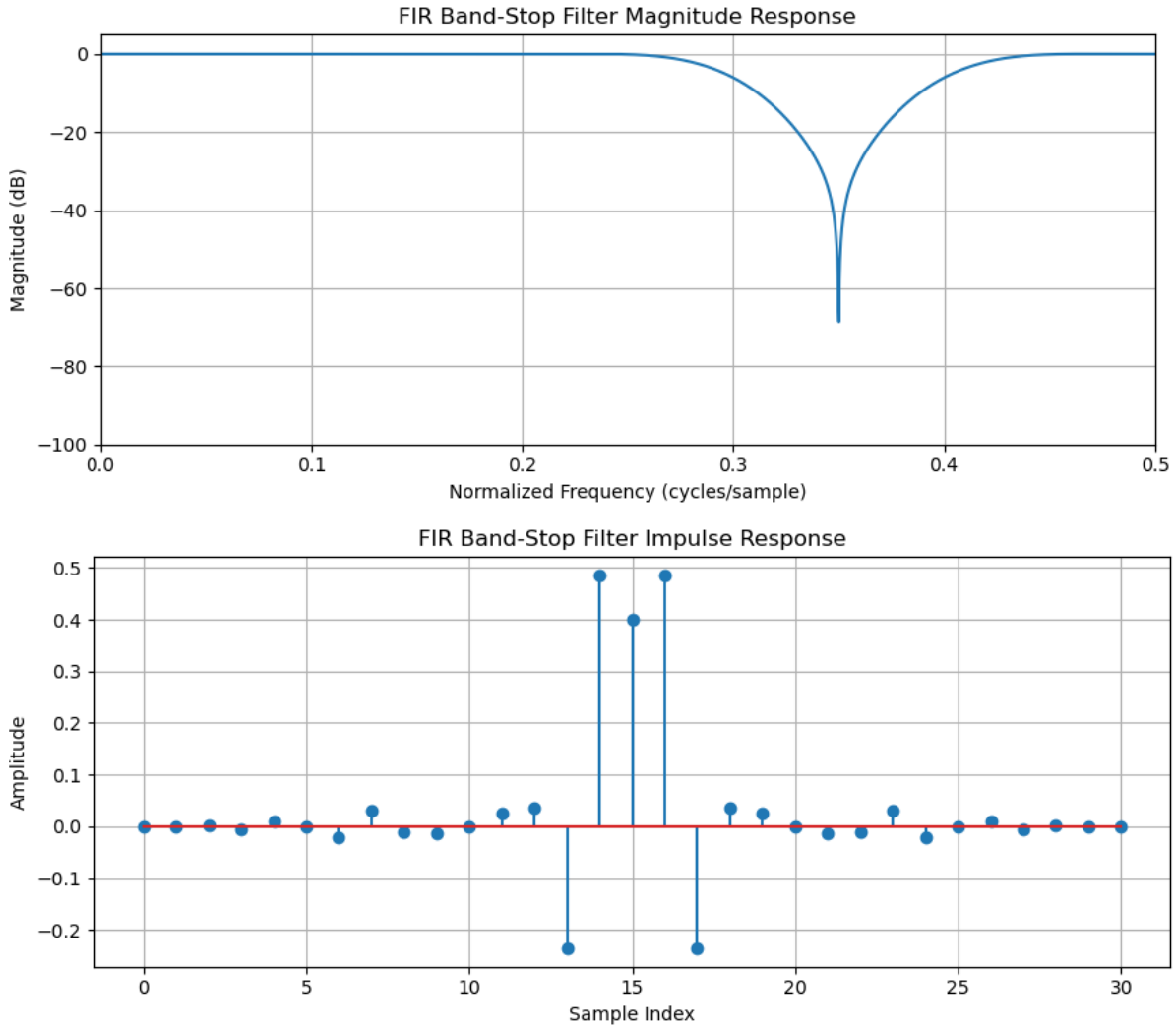
# Filter specifications
cutoff_frequency1 = 3000 / 10000
cutoff_frequency2 = 6000 / 10000
filter_length = 31
window_type = 'hamming'

# Design and plot
filter_coefficients = design_fir_filter(
   cutoff_frequency1,
```

```
    cutoff_frequency2,
    filter_length,
    window_type
)

plot_filter_response(filter_coefficients)
```

Output:



Observation:

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- ✓ FIR band-stop filter coefficients were calculated using the windowing method with cutoff frequencies of 3 kHz and 6 kHz and a filter length of 31.
- ✓ The magnitude response was plotted to observe the frequency characteristics of the designed filter. However, the intended 3–6 kHz stopband cannot be achieved because the upper cutoff frequency exceeds the Nyquist frequency of 5 kHz.
- ✓ The impulse response contains 31 coefficients and is symmetrical about the centre sample.

Overall Conclusion:

- ➢ FIR filters were designed using the windowing method by applying the Hamming window to the ideal impulse response.
- ➢ Low-pass and high-pass FIR filters were implemented, and their magnitude and impulse responses were plotted to study their frequency characteristics.
- ➢ Band-pass and band-stop filter coefficients were calculated using the specified cutoff frequencies, and their responses were plotted for analysis.